Date: 2026-07-30
Request ID: 100048830

Katanax		
Analyte(s)	Cr, Cu, Mg, Si, Ti	
SOP#		
Acid Cocktail	30 mL conc. HCl	
sample(s)	**weight (g)**	**volume (mL)**
200128736_1	0.3097	250
200128736_2	0.3059	250
200128737_1	0.3057	250
200128737_2	0.3023	250

Hotplate		
Analyte(s)	Na	
SOP#		
Cocktail	50 mL 1:1 HCl	
sample(s)	**weight (g)**	**volume (mL)**
200128736_1	0.5062	50
200128736_2	0.5043	50
200128737_1	0.5092	50
200128737_2	0.5053	50

Date: 2026-07-30
Request ID: 100048830

LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128736	16.6030	18.6492	18.5184	6.39	0.9361

Cr ICP analysis

sample	Dilution	conc. [mg/L]	Cr_ppm [dried]	%Cr [dried]
200128736_1	25/1	2.373	51166.55	5.12
200128736_2	25/1	2.322	50681.72	5.07

cr result: 50924.14 ppm
cr result: 5.09 %
cr oxide factor: 1.462
cr oxide result: 7.45 %
cr lot: *Manufacturer: Ricca, 45120423 exp: 2027-05-31*

Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm [dried]	%Cu [dried]
200128736_1	50/6	1.886	13555.18	1.36
200128736_2	50/6	1.856	13505.30	1.35

cu result: 13530.24 ppm
cu result: 1.35 %
cu oxide factor: 1.252
cu oxide result: 1.69 %
cu lot: *Manufacturer: Assurance, 28-83CUT exp: 2026-09-30*

Mg ICP analysis

sample	Dilution	conc. [mg/L]	Mg_ppm [dried]	%Mg [dried]
200128736_1	10/9	0.215	205.69	0.02
200128736_2	10/9	0.215	208.24	0.02

mg result: 206.97 ppm
mg result: 0.02 %
mg lot: *Manufacturer: Assurance, 28-136MGY exp: 2027-01-30*

Na ICP analysis

sample	Dilution	conc. [mg/L]	Na_ppm [dried]	%Na [dried]
200128736_1	25/1	0.864	2279.24	0.23
200128736_2	25/1	0.877	2322.25	0.23

na result: 2300.75 ppm
na result: 0.23 %
na lot: *Manufacturer: Assurance, 27-159NAY exp: 2026-08-30*

Si ICP analysis

sample	Dilution	conc. [mg/L]	Si_ppm [dried]	%Si [dried]
200128736_1	250/250	1.095	944.28	0.09
200128736_2	250/250	1.299	1133.68	0.11

si result: 1038.98 ppm

si result: 0.10 %

si lot: *Manufacturer: Assurance, 28-87SIX exp: 2026-08-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm [dried]	%Ti [dried]
200128736_1	10/9	0.058	55.57	0.01
200128736_2	10/9	0.044	42.68	0.00

ti result: 49.13 ppm

ti result: 0.00 %

ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s):

Date: 2026-07-30
Request ID: 100048830

LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128737	18.6915	20.8111	20.7101	4.77	0.9523

Cr ICP analysis

sample	Dilution	conc. [mg/L]	Cr_ppm [dried]	%Cr [dried]
200128737_1	25/0.77	2.105	58697.33	5.87
200128737_2	25/0.77	2.147	60532.25	6.05

cr result: 59614.79 ppm
cr result: 5.96 %
cr oxide factor: 1.462
cr oxide result: 8.72 %
cr lot: *Manufacturer: Ricca, 45120423 exp: 2027-05-31*

Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm [dried]	%Cu [dried]
200128737_1	10/1	1.606	13786.64	1.38
200128737_2	10/1	1.729	15016.31	1.50

cu result: 14401.48 ppm
cu result: 1.44 %
cu oxide factor: 1.252
cu oxide result: 1.80 %
cu lot: *Manufacturer: Assurance, 28-83CUT exp: 2026-09-30*

Mg ICP analysis

sample	Dilution	conc. [mg/L]	Mg_ppm [dried]	%Mg [dried]
200128737_1	10/9	0.124	118.31	0.01
200128737_2	10/9	0.125	120.61	0.01

mg result: 119.46 ppm
mg result: 0.01 %
mg lot: *Manufacturer: Assurance, 28-136MGY exp: 2027-01-30*

Na ICP analysis

sample	Dilution	conc. [mg/L]	Na_ppm [dried]	%Na [dried]
200128737_1	25/1	0.810	2086.61	0.21
200128737_2	25/1	0.812	2107.91	0.21

na result: 2097.26 ppm
na result: 0.21 %
na lot: *Manufacturer: Assurance, 27-159NAY exp: 2026-08-30*

Si ICP analysis

sample	Dilution	conc. [mg/L]	Si_ppm [dried]	%Si [dried]
200128737_1	1/1	1.657	1422.46	0.14
200128737_2	1/1	1.443	1252.63	0.13

si result: 1337.54 ppm
si result: 0.13 %
si lot: *Manufacturer: Assurance, 28-87SIX exp: 2026-08-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm [dried]	%Ti [dried]
200128737_1	10/9	0.037	34.83	0.00
200128737_2	10/9	0.032	30.88	0.00

ti result: 32.85 ppm
ti result: 0.00 %
ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s):

ppm M+ = [conc.][volume][dilution]/[weight]

%M+ = [conc.][volume][dilution]/([weight]*10,000)

%MO = [oxide factor]*%M+

A = crucible
B = crucible + sample
C = crucible + sample after drying

%LOI = ([B-A]-[C-A])/(B-A)*100%

ppm M+ [dried] = ppm M+/([1-(%LOI)/100])

%M+ [dried] = %M+/([1-(%LOI)/100])

%MO [dried] = %MO/([1-(%LOI)/100])